A3-PS003-TC001

PS-003: Federated Traffic-Flow Prediction with Communication-Cost Evaluation

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Problem & Motivation

PS-003: forecast traffic flow without streaming every raw sensor reading to one place

Bandwidth-heavy centralization

Roadside sensors and edge cameras must continuously stream raw data to a central cloud, consuming high cellular bandwidth.

Communication cost is not quantified

The trade-off between prediction accuracy and bytes communicated is rarely measured side by side.

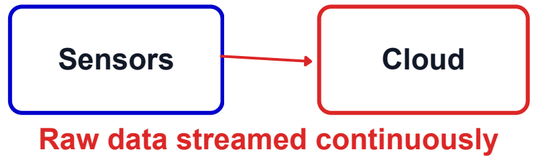
Privacy and data locality

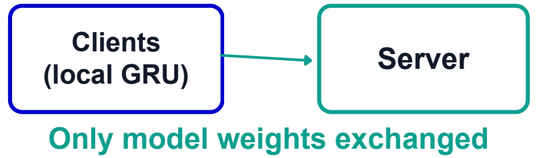
Centralizing raw travel logs concentrates sensitive data. Federated training keeps raw data at simulated clients; no formal privacy guarantee is claimed.

Non-IID traffic across intersections

Peak hours, road types and sensor locations differ per intersection, so one global model can drift or converge slowly.

Centralized vs federated data movement





Raw data remains at simulated clients

Goal: compare federated and centralized forecasting, and report MAE / RMSE against bytes sent per training round.

Background: Traffic Forecasting + Federated Learning

Concepts needed to understand the problem and the proposed approach

Traffic-flow prediction

- Predict future traffic flow from past sensor readings over time
- Input: a window of recent readings per sensor
- Output: the next value(s) over a short horizon [5]

Federated Learning (FL) terms

Client	an intersection / sensor group with its own data
Local training	each client trains the model on its own data only
Server	collects client weights and aggregates them
FedAvg	averages client weights by local sample count [1]

FedAvg: weights, not raw data

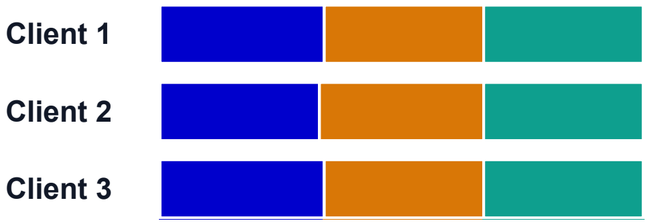
$$w(t+1) = \Sigma k \left(nk / n \right) \cdot wk(t+1)$$

nk = samples on client k, n = total

- Weights are far smaller than a continuous raw-data stream
- Raw readings never leave the client

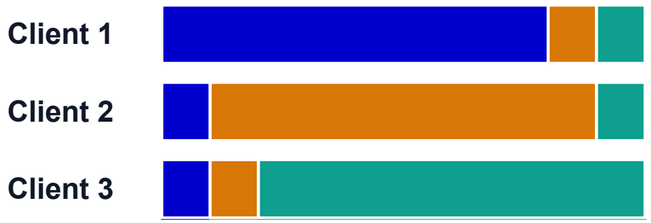
IID

Client traffic distributions are relatively similar



Non-IID

Clients have different traffic characteristics and peak patterns



Client setup

- **Simulated clients** = intersections (executed run: 3 clients)
- **Each client** gets its own slice of the training data
- **IID:** windows spread evenly. **Non-IID:** different sensor groups / regimes [4]
- Colours = traffic patterns (conceptual, not data)

Dataset & Literature Review

The dataset used by the prototype, and the most relevant existing work

Dataset: Synthetic Traffic Data (Prototype / Demo)

- **Target:** traffic flow = vehicle count per 15-min interval (not speed, not occupancy)
- **Size:** 2,688 intervals (28 days × 96 per day)
- **Split:** chronological 80% train (~2,150) / 20% test (~538); no random shuffle
- **Clients:** simulated intersection profiles (normal bimodal, morning-peak, evening-peak, arterial, residential)
- **Purpose:** test the federated pipeline end to end; not real-city measurements

Data layout (schematic)

time step (15 min)	client 1	client 2	client 3
t	*x1(t)*	*x2(t)*	*x3(t)*
t + 1	*x1(t+1)*	*x2(t+1)*	*x3(t+1)*

Future / alternative public datasets (not used in current experiments): METR-LA, PeMS

Existing work and standards

[1] McMahan et al.: FedAvg, communication-efficient training from decentralized data

[2] Liu et al.: FedGRU, federated GRU-based traffic-flow prediction

[3] Li et al.: FedProx, a proximal term for heterogeneous clients

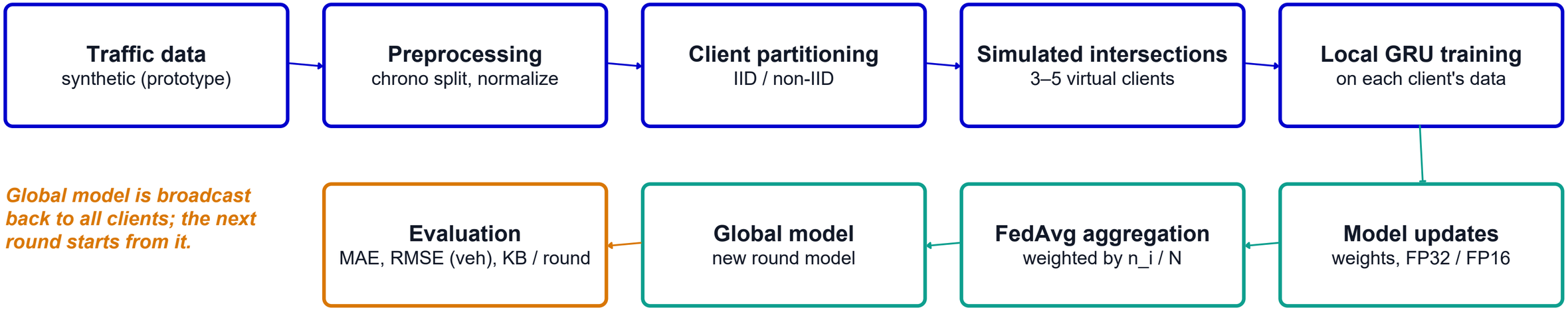
[4] Hsieh et al.: how non-IID data degrades decentralized learning

[5] Li et al.: DCRNN, source of the METR-LA benchmark (future / alternative dataset)

[6] IEEE 3652.1-2020: federated ML architectural framework guide. No IETF RFC applies directly.

Proposed Solution & Workflow

Simulated traffic clients train a GRU locally; the server only aggregates model updates



Compact GRU forecaster

- PyTorch GRU: 1 GRU layer + Linear head, 3,393 parameters
- Input: sliding window W = 12 past 15-min intervals
- Output: next-interval traffic flow (vehicles)

Centralized baseline (comparison reference)

- Same GRU trained on all pooled training data
- Same test set and metrics as the federated runs
- Compared against FedAvg IID and non-IID

Raw traffic data remains at simulated clients during federated training; only model updates / weights are exchanged. This prototype does not implement formal differential privacy or secure aggregation.

Communication Cost & Optimization

How bytes per round are computed, and the optimizations the prototype supports

How communication cost is calculated

$$C(\text{round}) \approx P \times b \times 2 \times K$$

- **P:** trainable parameters (GRU: 3,393)
- **b:** bytes per parameter: FP32 = 4, FP16 = 2
- **2:** client → server upload + server → client download
- **K:** number of clients

Executed run (K = 3, FP32)
3,393 × 4 × 2 × 3 = 81,432 B ≈ 79.52 KB / round
5 rounds → 0.388 MB total (measured)
FP16 (measured): 39.76 KB / round, about 50% fewer bytes

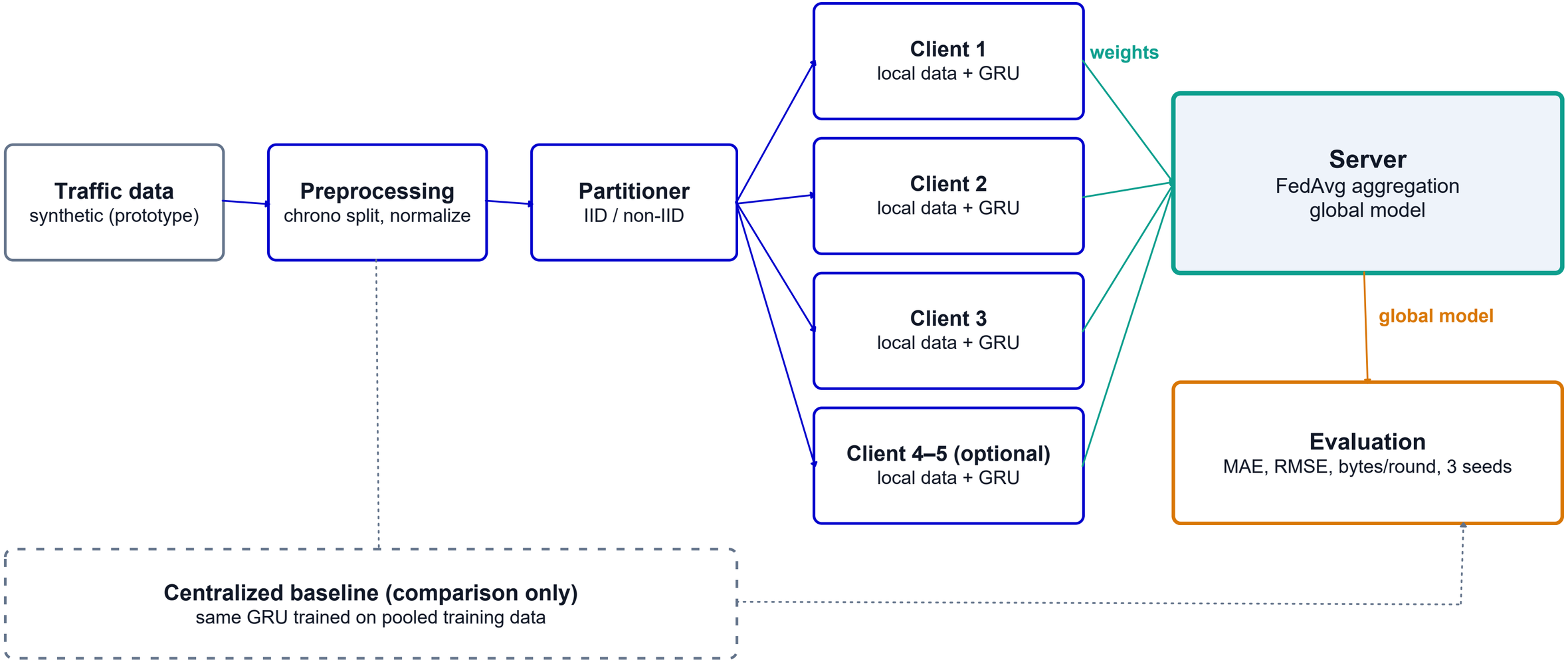
Reported in bytes, KB (÷1024) and MB. Total = per-round cost × rounds.

Aspect	Baseline FedAvg	Optimized FL
Update size	FP32 updates, 4 bytes per parameter	FP16 updates, 2 bytes per parameter (measured 39.76 vs 79.52 KB per round)
Local training	fixed local epochs and learning rate	tuned local epochs, LR decay, early stopping
Non-IID handling	one global normalization	per-client normalization where appropriate
Data split	random split risks leakage	chronological train/test split

Measured: FP16 39.76 KB vs FP32 79.52 KB per round (about 50% less). Error stays comparable; no accuracy gain is claimed.

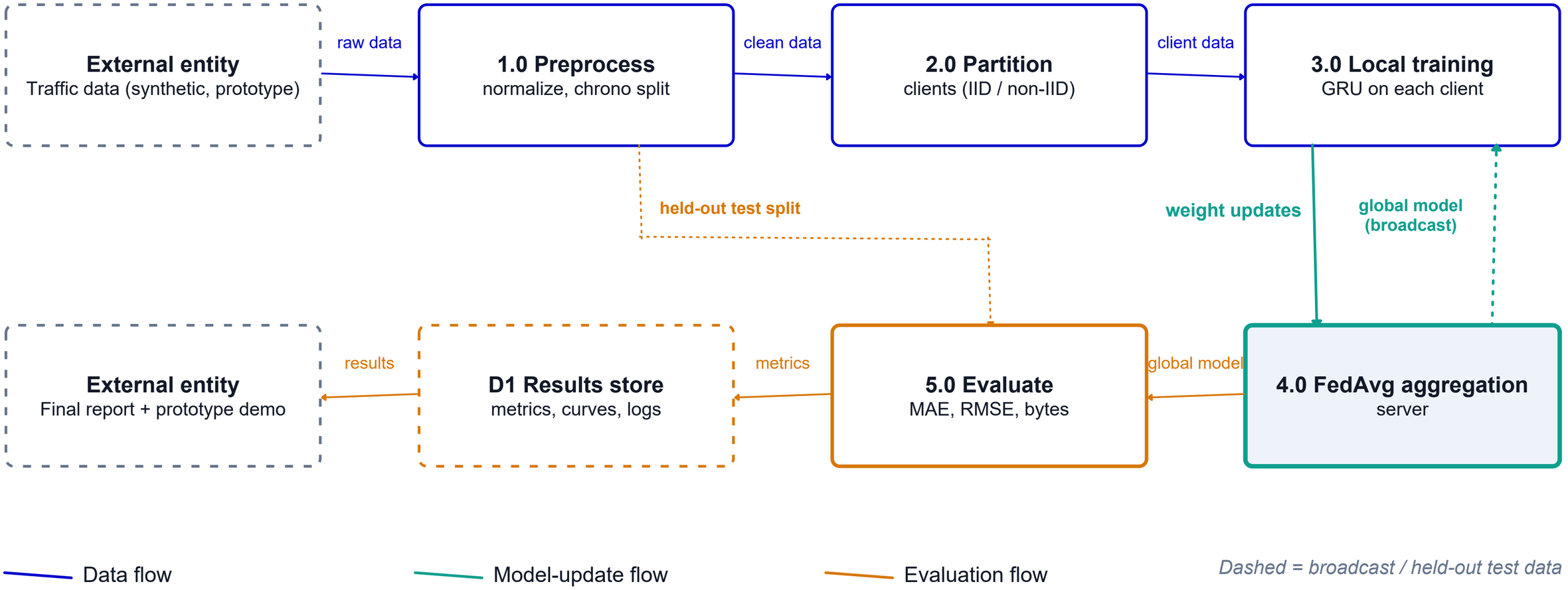
System Architecture

Clients train locally and upload weights; the server aggregates and broadcasts the global model back



Data Flow Diagram / Federated Training Flow

How data, model updates and evaluation results move through the simulator



Experimental Design

Four executed configurations and the settings of the prototype runs

Experiment configurations

Method	Setting	Precision	Status
Centralized baseline	pooled training data	n/a	✓ Executed
FedAvg — IID	IID clients	FP32	✓ Executed
FedAvg — Non-IID	non-IID clients	FP32	✓ Executed
Optimized FL	Non-IID, FP16 updates	FP16	✓ Executed

All four main experiments executed; results on the next slide.

Executed run settings

- 3 simulated clients, IID and Non-IID modes
- 5 federated rounds, local epochs E = 1
- Window W = 12, 80/20 chronological split
- FP32 and FP16 updates, FedAvg aggregation
- Also supported: learning rate, batch size, random seed

3-seed robustness

Seed 42 **Seed 43** **Seed 44**

Status: Planned. Seeds 42, 43, 44 have not been run, so no mean ± std is reported.

MAE

mean absolute error, in vehicles (lower is better)

RMSE

root mean squared error, penalizes large misses

Comm. / round

upload + download bytes, from the formula

Convergence

rounds until error stops improving

Experimental Results & Comparison

Measured by the prototype: 3 simulated clients, 5 federated rounds

Method	Data Setting	MAE	RMSE	Communication / Round
Centralized	Combined	16.83 veh	23.67 veh	N/A (Centralized Cloud)
FedAvg IID	IID	12.49 veh	15.95 veh	79.52 KB (FP32)
FedAvg Non-IID	NON-IID	17.09 veh	24.43 veh	79.52 KB (FP32)
Optimized FL	NON-IID	16.77 veh	23.94 veh	39.76 KB (FP16)

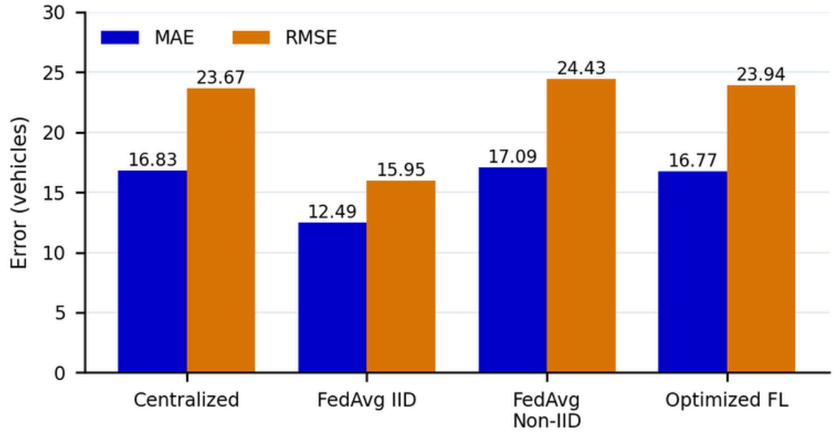
- FP16 cuts model-update traffic from 79.52 to 39.76 KB per round (about 50%) while error stays comparable
- Centralized communication is N/A: no client-server model exchange
- 3-seed robustness (seeds 42, 43, 44) is planned, not run: no mean ± std

Measured data heterogeneity (3 simulated clients)

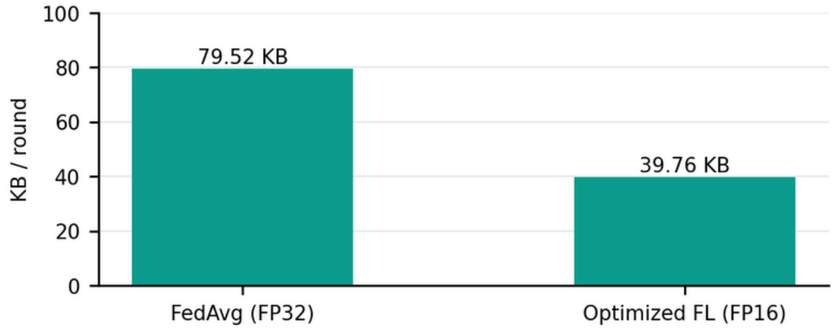
Client	IID mean	IID std	Non-IID mean	Non-IID std	Non-IID peak
1	118.2	67.4	125.8	60.1	266.4
2	117.5	68.5	101.9	62.1	338.6
3	119.8	68.0	128.8	77.7	340.6

Flow in vehicles per 15 min.

MAE and RMSE by method (vehicles, lower is better)



Communication per round (KB)



Conclusion, Scope & References

Conclusion

Our experiments demonstrate that federated traffic-flow prediction can maintain useful prediction accuracy while reducing model-update communication. Using FP16 reduces the measured communication cost from 79.52 KB/round to 39.76 KB/round, approximately a 50% reduction, while the measured prediction errors remain comparable.

- Centralized: MAE 16.83, RMSE 23.67 veh. FedAvg IID: MAE 12.49, RMSE 15.95 veh
- FedAvg Non-IID: MAE 17.09, RMSE 24.43 veh. Optimized FL: MAE 16.77, RMSE 23.94 veh
- Communication: FP32 79.52 KB/round, FP16 39.76 KB/round (about 50% less)

Scope & limitations

- Simulated clients, synthetic prototype dataset
- No live camera deployment
- No differential privacy or secure aggregation
- 3-seed robustness (seeds 42–44) not yet run

References

[1] H. B. McMahan et al. “Communication-Efficient Learning of Deep Networks from Decentralized Data.” In: Proc. AISTATS, PMLR 54 (2017), pp. 1273–1282.

[2] Y. Liu et al. “Privacy-Preserving Traffic Flow Prediction: A Federated Learning Approach.” In: IEEE Internet of Things Journal 7.8 (2020), pp. 7751–7763.

[3] T. Li et al. “Federated Optimization in Heterogeneous Networks.” In: Proc. MLSys (2020).

[4] K. Hsieh et al. “The Non-IID Data Quagmire of Decentralized Machine Learning.” In: Proc. ICML (2020).

[5] Y. Li et al. “Diffusion Convolutional Recurrent Neural Network: Data-Driven Traffic Forecasting.” In: Proc. ICLR (2018).

[6] IEEE Std 3652.1-2020, guide for the architectural framework of federated machine learning.

Thank You for Your Attention!

Team Name | A3-PS003-TC001